

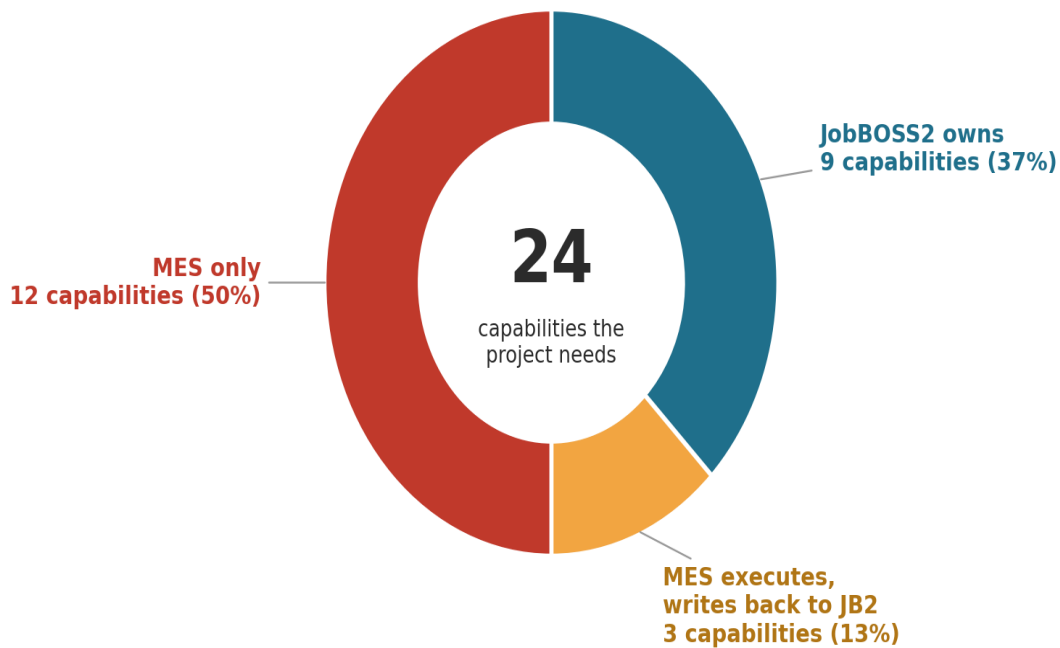
JobBOSS² vs. Atlas MES — Capability Split

Companion to the MES Design Document · JB2 API facts verified against the live OpenAPI spec (api-jb2.integrations.ecimanufacturing.com, 2026-07-10)

Rule of thumb: JobBOSS² owns **money and orders**; the MES owns **execution and evidence**.

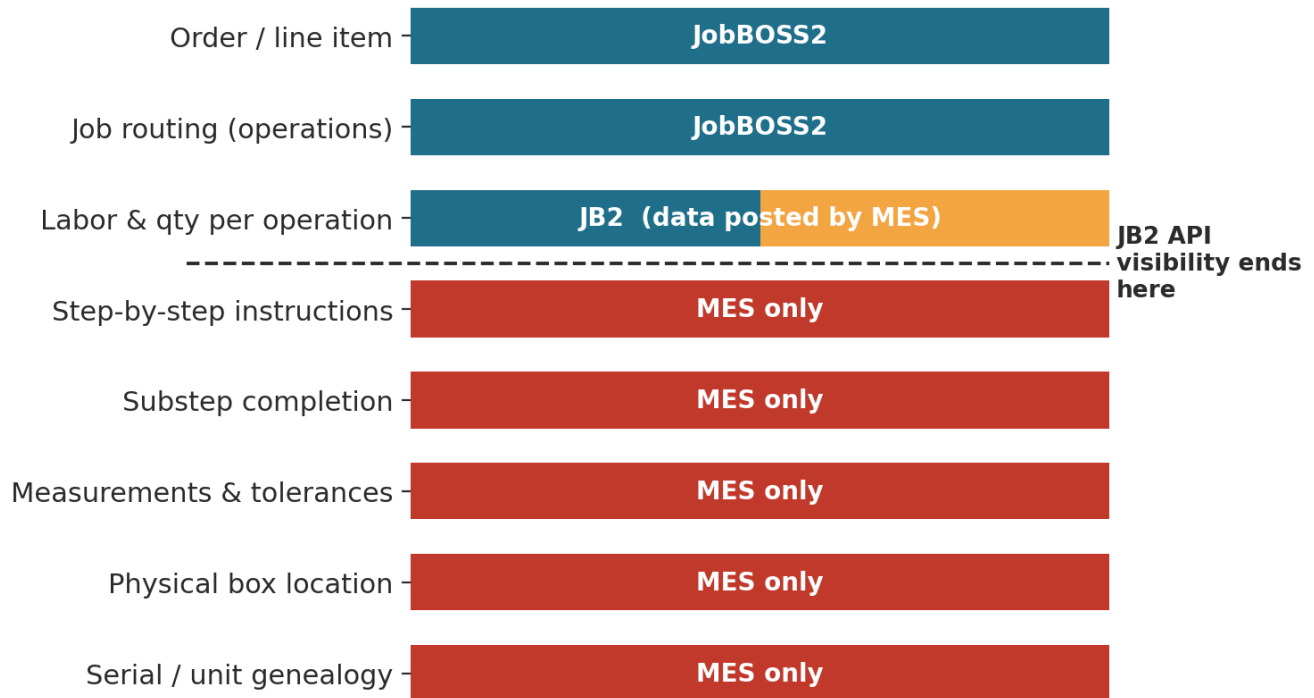
Who owns the 24 capabilities this project requires

Half of what the project needs does not exist in JobBOSS² at any level. Another 13% exists in JB2 but only works because the MES posts the data.



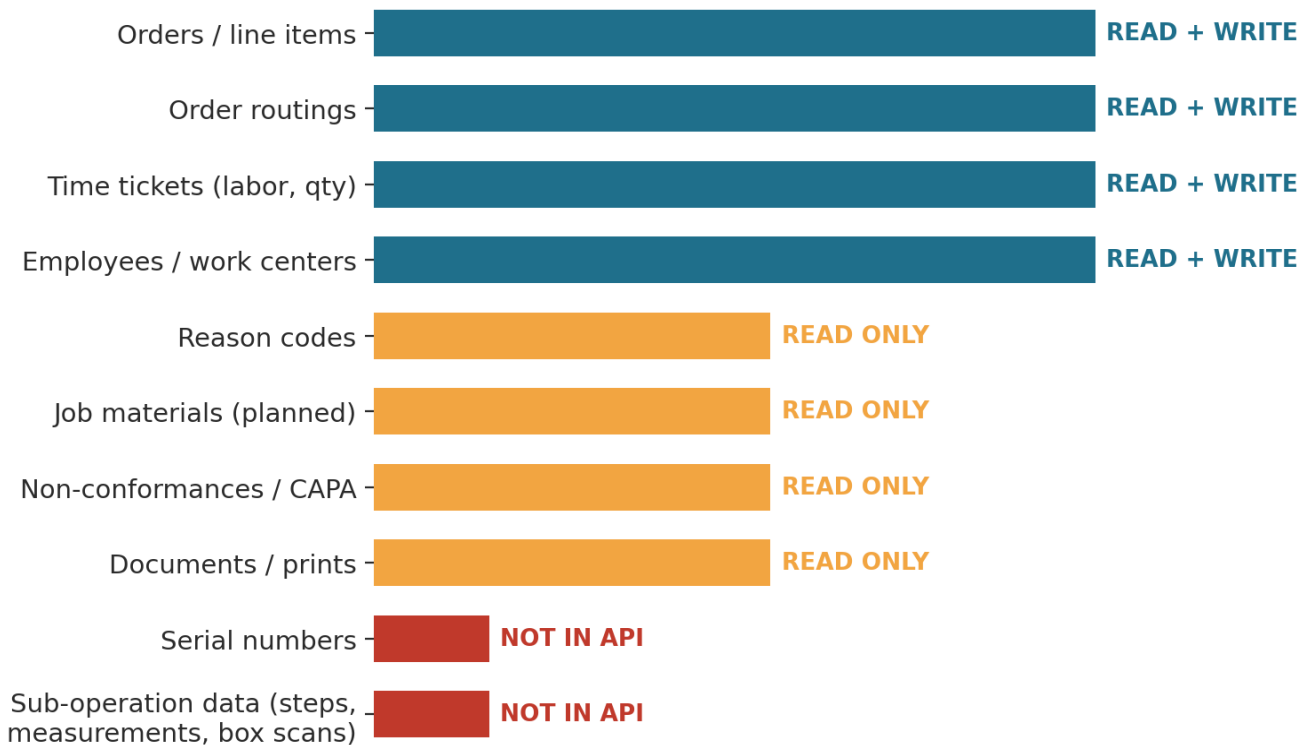
How deep each system can see

JB2's data model stops at the routing operation. Everything below that line — the layer operators actually work in — is invisible to JB2 and is the MES's reason to exist.



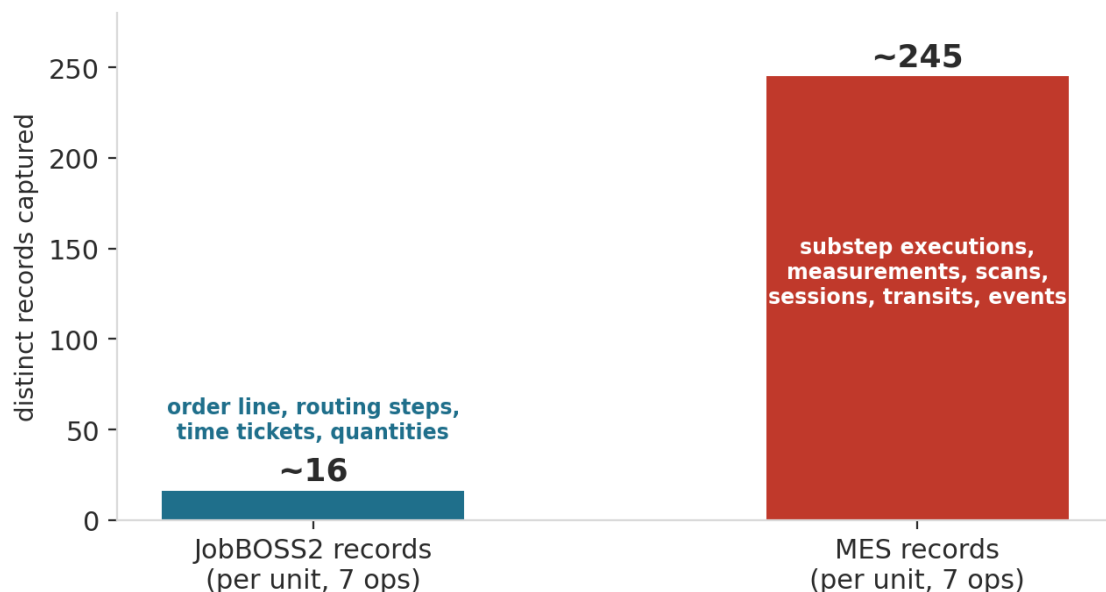
JB2 API access to what the project needs

Verified endpoint by endpoint: full read/write where it matters for costing (labor, quantities), read-only for materials and quality, and no API surface at all for serials or anything below the operation.



Same unit, two very different records

Illustrative record counts for one pistol passing seven operations: JB2 ends up with roughly 16 records (order line, routing steps, time tickets). The MES captures ~245 (every substep, measurement, scan, session and transit) — about 15× the granularity, which is what makes first-pass-yield and rework analytics possible.



Area-by-area split

Area	JobBOSS ² already handles	MES must handle
Orders	Order entry, line items, parts, quantities, due dates, revisions (full read API, incremental sync via lastModDate)	Auto-create tracked work orders + execution plans per unit
Routing	Authoring routings: steps, operation codes, work centers, time estimates	Bind steps to versioned instructions; freeze plan per order
Work instructions	One text line per operation; no media, no versions	Full library: steps/substeps, images, tolerances, versioning, PDF guides
Labor / time	Labor tickets per employee/job/op: start-end, setup vs cycle (MES posts automatically)	Substep timestamps, pause reasons, rework vs first-pass time, queue time
Quantities	Pieces finished/scrapped per operation; routing actuals via PATCH	Per-unit outcomes in a batch; remake genealogy
Operation status	'Job at operation N', current work center	Real-time physical box location + full scan history
Scrap reasons	Flat reason-code list on time tickets	Failure taxonomy, photos, dispositions, rework loops, first-pass flag
Materials	Planned material per step (READ-ONLY - no API to post actuals)	Actual usage, deltas vs plan, substitutions - sole record
Quality	NC/CAPA modules exist but are READ-ONLY via API	All measurements, tolerances, inspections, sign-offs - sole record
Serial numbers	Not in the API	Serial capture + full unit traceability
People	Employee master; payroll attendance	Badge auth, roles, who-did-what, lead overrides
Visibility	Office reports, ShopView, APS schedule	Live pipeline dashboard, first-pass vs rework, bottleneck analytics
Money	Costing, invoicing, purchasing - fed by MES write-backs	Nothing. MES never touches money.

The one-sentence version

JobBOSS² knows **what to build, in what order, and what it cost**. The MES knows — and JB2 has no way to know — **exactly what happened to each physical unit**: every instruction shown, box scanned, minute worked, dimension measured, failure dispositioned, and rework loop taken.

Record-count chart is illustrative (typical 7-operation build, ~5 steps/op, ~4 substeps/step). Capability counts from the design document matrix C1-C24. API access verified 2026-07-10.